Akoizo Original Practice - Form 3 - Answer key

Division B | Botany | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) The Casparian strip particularly restricts…

Unit: Roots, stems, and leaves.
Worked-case data: A leaf closes its stomata during dry conditions. Predict one benefit and one cost.

A. Unregulated apoplastic entry

B. All photosynthesis

C. Every phloem tube

D. All root growth

Correct: A

2. (2 points) Ferns reproduce through…

Unit: Major plant groups and comparative identification.
Worked-case data: A specimen has parallel leaf venation, floral parts in threes, and scattered stem bundles. Identify the supported broad group.

A. Spores with a gametophyte stage

B. Flowers and fruit universally

C. Only seeds

D. No meiosis

Correct: A

3. (2 points) A fungal structure on a lesion is an example of…

Unit: Plant ecology, nutrient deficiencies, and disease.
Worked-case data: Older leaves yellow first, but soil is dry and no tissue analysis exists. Is nutrient deficiency established?

A. A sign

B. Only a symptom

C. A root system

D. A light-response plateau

Correct: A

4. (2 points) Do all plant cells have many chloroplasts?

Unit: Plant cells, tissues, and vascular organization.
Worked-case data: A mature leaf exports sugar to a growing root. Identify source, sink, and transport tissue.

A. Only every dead xylem cell

B. No

C. Yes

D. Only every root cell

Correct: B

5. (2 points) Pollination and fertilization are…

Unit: Life cycles, flowers, seeds, and biotechnology.
Worked-case data: Track ploidy in the standard angiosperm pathway: diploid sporophyte, spore, gametophyte, gamete, zygote.

A. Both meiosis

B. Different events

C. Synonyms

D. Both always seed dispersal

Correct: B

6. (2 points) Oldest known fossil equals exact origin time?

Unit: Plant evolution, fossils, and diversity.
Worked-case data: Two modern lineages share a vascular-tissue ancestor. Does one living lineage have to be the ancestor of the other?

A. Only for pollen

B. Not necessarily

C. Always

D. Only if the fossil is large

Correct: B

7. (2 points) A herbarium primarily preserves…

Unit: Botany, evidence, and ecosystem importance.
Worked-case data: A plant community fixes 100 carbon units and its producers respire 40 units over the same interval. Calculate NPP.

A. Every soil variable

B. A full evolutionary tree automatically

C. Specimens and collection evidence

D. A continuous gas-exchange record

Correct: C

8. (2 points) Hormone effects depend on…

Unit: Water transport, storage, hormones, and growth.
Worked-case data: Soil water potential is -0.2 MPa, root -0.5 MPa, leaf -1.0 MPa. Predict the idealized direction.

A. Only leaf color

B. No concentration effects

C. Tissue and context

D. Only the hormone name

Correct: C

9. (4 points) Explain “Chloroplast” in your own words.

Unit: Photosynthesis and plant energy.
Worked-case data: Low light: 5 bubbles per minute. Medium light: 10 bubbles per minute. High light: 10 bubbles per minute at the same temperature and carbon-dioxide supply.
Separate investigation observations:
Low light: 5 bubbles per minute.
Medium light: 10 bubbles per minute.
High light: 10 bubbles per minute at the same temperature and carbon-dioxide supply.

answer: Plant cell structure where photosynthesis occurs. (4 points).

10. (2 points) Vegetative propagation commonly helps preserve…

Unit: Horticulture, aquatic systems, medicine, and materials.
Worked-case data: Two irrigation treatments produce the same growth index, but one uses twice the water. What else is needed to choose a management approach?

A. No traits

B. Every environmental effect

C. A selected genotype

D. Maximum new genetic variation necessarily

Correct: C

11. (4 points) Compare apoplast and symplast.

Unit: Roots, stems, and leaves.
Worked-case data: A leaf closes its stomata during dry conditions. Predict one benefit and one cost.

answer: Apoplast follows walls/extracellular spaces; symplast follows connected cell interiors. (4 points).

12. (4 points) Name two major seed-plant innovations.

Unit: Major plant groups and comparative identification.
Worked-case data: A specimen has parallel leaf venation, floral parts in threes, and scattered stem bundles. Identify the supported broad group.

answer: Seeds protect embryos; pollen and reduced gametophytes alter reproductive dispersal and dependence on external water. (4 points).

13. (4 points) Distinguish symptom and sign.

Unit: Plant ecology, nutrient deficiencies, and disease.
Worked-case data: Older leaves yellow first, but soil is dry and no tissue analysis exists. Is nutrient deficiency established?

answer: A symptom is the plant response; a sign is observable evidence of the agent or its structures. (4 points).

14. (4 points) What is an apical meristem’s role?

Unit: Plant cells, tissues, and vascular organization.
Worked-case data: A mature leaf exports sugar to a growing root. Identify source, sink, and transport tissue.

answer: It supplies cells contributing to primary growth in length and associated new tissues. (4 points).

15. (4 points) What commonly forms triploid endosperm?

Unit: Life cycles, flowers, seeds, and biotechnology.
Worked-case data: Track ploidy in the standard angiosperm pathway: diploid sporophyte, spore, gametophyte, gamete, zygote.

answer: A sperm joins the central cell containing two maternal polar-nuclear contributions in the standard arrangement. (4 points).

16. (4 points) Why can older leaves show deficiency first?

Unit: Plant ecology, nutrient deficiencies, and disease.
Worked-case data: Older leaves yellow first, but soil is dry and no tissue analysis exists. Is nutrient deficiency established?

answer: Some mobile nutrients are redistributed to new growth, leaving older tissues depleted. (4 points).

17. (4 points) Why are lignified xylem walls useful?

Unit: Plant cells, tissues, and vascular organization.
Worked-case data: A mature leaf exports sugar to a growing root. Identify source, sink, and transport tissue.

answer: They support the plant and resist collapse under the tension of water transport. (4 points).

18. (4 points) Why is a breeding method not an outcome guarantee?

Unit: Life cycles, flowers, seeds, and biotechnology.
Worked-case data: Track ploidy in the standard angiosperm pathway: diploid sporophyte, spore, gametophyte, gamete, zygote.

answer: Effects depend on the specific trait, organism, environment, and evaluation; method labels alone do not establish performance or ecological consequences. (4 points).